CS2 Week 3

State Feedback, Pole Placement, LQR

Exercise Classes

- Shared exercise with Haixiao and Kissan
- Theory recap
- Notes available on Kissan's website: https://kissanv.github.io

& on Haixiao's Polybox: https://polybox.ethz.ch/index.php/s/AReA9YBeJQKxBDn

- Quizzes, Examples & Old exam questions

Kissan's Website

Haixiao's Polybox

Recap Last Week

Quiz 1

When is a system said to be reachable?

A. If there exists a signal, which can take a system with initially $x = 0$ to any arbitrary state x_{final}

B. If, for any initial condition x_0, this initial condition can be reconstructed uniquely based only on the knowledge of the input and output signal over a finite time interval

C. If, the Reachability matrix has rank n

D. When the input tries hard enough

Quiz 2

Which modes of the system are reachable?

$$\tilde{A} = \begin{bmatrix} j & 0 & 0 & 0 \\ 0 & -j & 0 & 0 \\ 0 & 0 & j & 0 \\ 0 & 0 & 0 & -j \end{bmatrix}, \quad \tilde{B} = \begin{bmatrix} 0.707j \\ -0.707j \\ 0 \\ 0 \end{bmatrix}$$

$$\tilde{C} = \begin{bmatrix} 0 & 0 & 0.707 & 0.707 \end{bmatrix}, \quad \tilde{D} = [0].$$

A. 1^{st} mode

B. 2^{nd} mode

B. 3^{rd} mode

B. 4^{th} mode

- PBH test for detectability:

$$\operatorname{rank}\begin{bmatrix}\lambda I - A \\ C\end{bmatrix} = n \quad \forall \lambda \in \mathbb{C} \text{ with } \mathrm{Re}(\lambda) \geq 0 \text{ in CT or } |\lambda| \geq 1 \text{ in DT.}$$

Quiz 3

Is the following system detectable?

$$A = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix} \qquad C = \begin{bmatrix} 1 & -1 \end{bmatrix}$$

Course Schedule

Subject	Week
Intro – Discrete Time Systems	1
Diagonalization, Modal Coordinates, Contr/Obs.	2
State Feedback, Pole Placement, LQR	3
State Estimation, Luenberger Obs., Kalman Filter	4
03/17 Separation Principle, LQG, Stability margins	5
Intro to MIMO Systems, Transmission Zeros	6
Nonlinear Systems	7
break	8
Discrete-time Unconstrained Optimal Control	9
Convex Optimization	10
Model Predictive Control	11
Practical Model Predictive Control	12
Robust Model Predictive Control	13
Implementation Methods	14

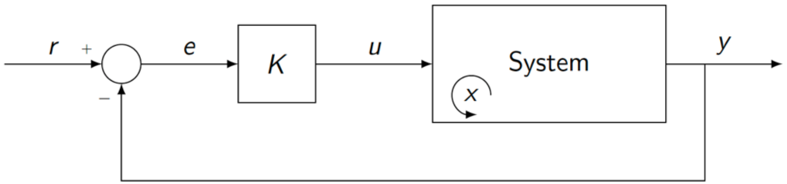
Today

1. State Feedback
2. Pole Placement
3. LQR

1. State Feedback

Feedback Control

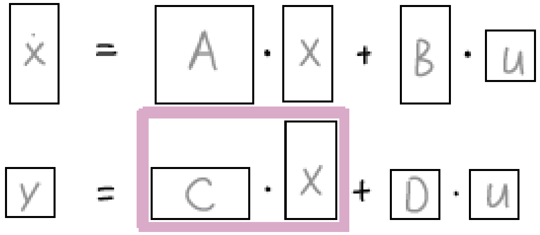
Let's take a step back and look at how we did feedback until now (CS1):



- We assumed that we could only measure the output y
- For a system in state-space form: $$\dot{x}(t) = Ax(t) + Bu(t)$$ $$y(t) = Cx(t) + Du(t)$$

we were able to measure y only not x

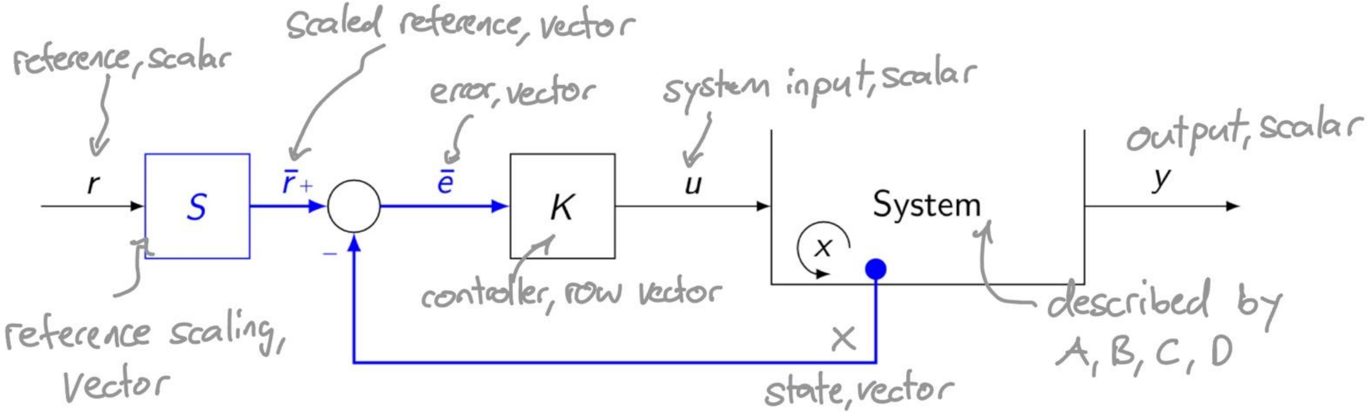
- look at the dimensions: example of a typical 2nd order system:
- the output has lees dimension than x



⇒ output y holds less information than states x

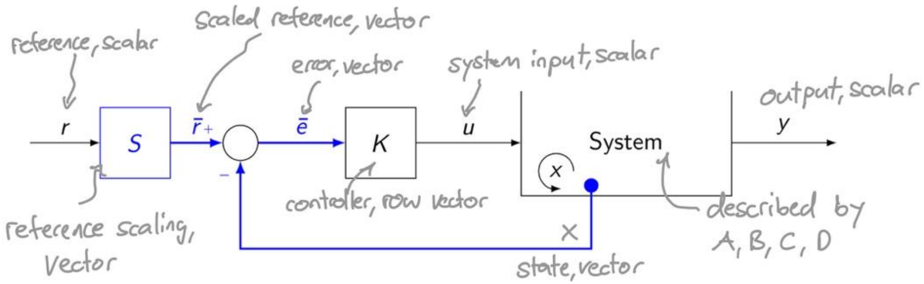
State Feedback

- now, assume that we know the states x → smarter to directly feed back x
- the control architecture would look like this:



- How does this affect the equations for our closed loop system?

State Feedback



- from the block diagram we read:

$$\left.\begin{aligned} \bar{r}_{+} &= Sr \\ \left.\begin{aligned} \bar{e} &= \bar{r}_{+} - x \\ u &= K\bar{e} \end{aligned}\right\} & \ \bar{e} = Sr - x \end{aligned}\right\} u = K(Sr - x) = -Kx + KSr$$

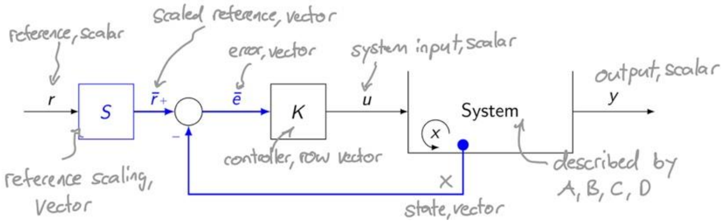
- insert this in our state-space-model $A, B, C, D = 0$

$$\left.\begin{aligned} \dot{x} &= (A - BK)x + B\bar{N}r \\ \dot{x} &= Ax - BKx + BKSr \\ \dot{x} &= Ax + Bu \end{aligned}\right\} \bar{N} := KS \longrightarrow \boxed{\begin{aligned} \dot{x} &= (A - BK)x + B\bar{N}r \\ y &= Cx \end{aligned}}$$

- so closing this loop gives us a new "closed loop A-Matrix" A_{cl}, where $A_{cl} = A - BK$

  $\Rightarrow$ through K we can influence the poles of our system!
  $\Rightarrow$ we can choose the system to behave how we want!

Steady State Error



$$\dot{x} = (A - BK)x + B\bar{N}r$$
$$y = Cx$$

- We need to cale the reference input to ensure consistent I/O dimensions.
- Idea scale the reference r such that that $e_{ss} = 0$

$\Rightarrow$ for that we need $G_{yr}^{cl}(0) = 1$ why?

Assuming the closed-loop system is stable, the steady-state response to a step input $r(t) = r_0$ is: $y_{ss} = G_{yr}^{cl}(0)r_0$

$$G_{yr}^{cl}(s) = C(sI - A + BK)^{-1}B\bar{N}$$
$$G_{yr}^{cl}(0) = C(-A + BK)^{-1}B\bar{N} = 1$$

$$\boxed{\Rightarrow\ \bar{N} = -[C(A - BK)^{-1}B]^{-1}}$$

The scaling vector S can be chosen in any such that $KS = \bar{N}$

2. Pole Placement

Pole Placement

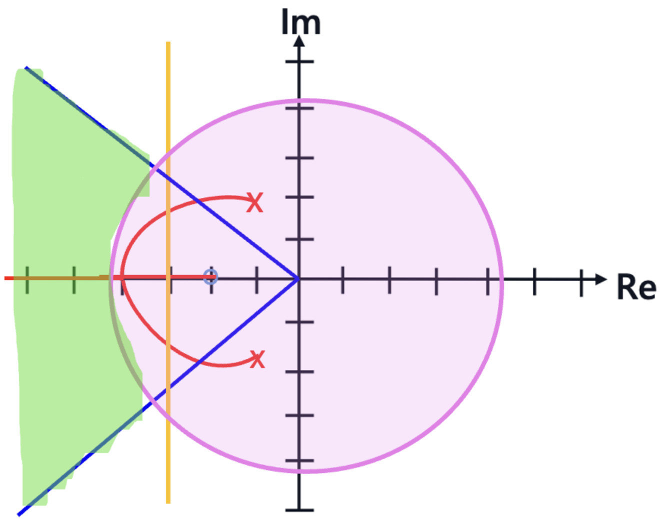
Specifications for 2nd order systems:

Settling Time: $T_d = \frac{1}{|\sigma|}\log\left(\frac{100}{d}\right)$

Peak Overshoot: $M_p = e^{\frac{\sigma\pi}{\omega}}$

Rise Time: $T_{100\%} = \frac{\frac{\pi}{2}-\varphi}{\omega} \approx \frac{\pi}{2\omega_n}$

Desired region



How do we construct K to have our poles in our desired positions?

- we are given the open loop system matrices A & B
- we know where our closed loop poles are supposed to be
- pole placement has a solution iff system is reachable

There are 2 main methods we will look at…

closed loop system:

$$\dot{x} = (A - BK)x + B\bar{N}r$$
$$y = Cx$$

Direct Method

recap: poles π_i are the eigenvalues of the A matrix

$\rightarrow \det(sI - A) = 0 = (s - \pi_1)(s - \pi_2) \dots$

1. Create the desired characteristic polynomial $p_{cl}^*(s) = \prod_{i=1}^{n}(s - \pi_i^*)$

2. Define the feedback matrix: $K = [k_1, k_2, \dots, k_n]$ with $n = \dim(A)$

3. Compute the closed-loop characteristic polynomial:
Characteristic polynomial: $p_{cl}(s) = \det(sI - A_{cl}) = \det(sI - A + BK)$

4. Coefficient comparison – We now have two expressions:
$p_{cl}(s)$ and $p_{cl}^*(s)$ Set: $p_{cl}(s) = p_{cl}^*(s)$

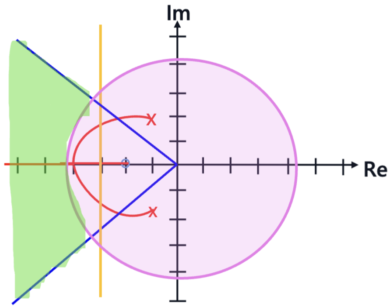
$\Rightarrow$ Solve for k_i via coefficient comparison.

Ackermann's Formula

$$\boxed{K = [0, \ldots, 0,1]R^{-1}p_{cl}^{*}(A)}$$

- R is the reachability matrix: $R = [B, AB, \ldots, A^{n-1}B]$
- is the desired characteristic polynomial evaluated at $s = A$.

Example (Nr.1 from PS)



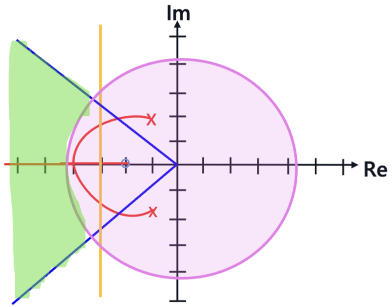
Let $\dot{x} = Ax + Bu, \quad A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

Design a state-feedback matrix K such that:

- No overshoot
- error decreases with e^{-3t}

Direct Method:

Example (Nr.1 from PS)



Let $\dot{x} = Ax + Bu, \quad A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

Design a state-feedback matrix K such that:

- No overshoot
- error decreases with e^{-3t}

Ackermann's Formula:

$$\boxed{K = [0, \dots, 0,1] R^{-1} p_{cl}^{*}(A)}$$

Reachable Canonical Form

Recall the reachable canonical form from CS1:

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 & \dots & 0 \\ 0 & 0 & 1 & 0 & \dots & 0 \\ \vdots & & & \ddots & & 1 \\ -a_0 & -a_1 & \dots & & & -a_{n-1} \end{bmatrix}, \qquad B = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}$$

$$C = \begin{bmatrix} c_0 & c_1 & \dots & & c_{n-1} \end{bmatrix}, \qquad D = [d];$$

With characteristic polynomial:

$$\varphi(s) = s^n + a_{n-1}s^{n-1} + \dots + a_0$$

$$\varphi(s) = s^n + a_{n-1}s^{n-1} + \ldots + a_0$$

Reachable Canonical Form

- If system is in reachable canonical form, and $K = [k_1, k_2, \ldots, k_n]$

$$A_{cl} = A - BK = \begin{bmatrix} 0 & 1 & 0 & 0 & \ldots & 0 \\ 0 & 0 & 1 & 0 & \ldots & 0 \\ \vdots & & & \ddots & & 1 \\ -a_0 - k_0 & -a_1 - k_1 & \ldots & & & -a_{n-1} - k_{n-1} \end{bmatrix}$$

- the closed loop characteristic polynomial is given by:

$$\varphi_{cl}(s) = s^n + (a_{n-1} + k_{n-1})s^{n-1} + \cdots + (a_1 + k_1)s + (a_0 + k_0)$$

- if the desired characteristic polynomial is given by: $\varphi^*_{cl}(s) = s^n + \alpha_{n-1}s^{n-1} + \cdots + \alpha_0$

→ easily compute k's by matching coefficients:

$$\boxed{k_i = \alpha_i - a_i \,,\ i = 0, \ldots, n-1}$$

Exam Question

$$k_i = \alpha_i - a_i \,,\; i = 0, \dots, n-1$$

α_i: coefficients of desired polyn.
a_i: coefficients of actual polyn.

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 & \dots & 0 \\ 0 & 0 & 1 & 0 & \dots & 0 \\ \vdots & & & \ddots & & 1 \\ -a_0 & -a_1 & \dots & & & -a_{n-1} \end{bmatrix}$$

$$\varphi_{cl}^*(s) = s^n + \alpha_{n-1}s^{n-1} + \dots + \alpha_0$$

Question 19 *Choose the correct answer. (1 Point)*

Consider the following LTI-system:

$$\dot{x}(t) = \begin{bmatrix} 0 & 1 \\ -1 & 1 \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t)$$

For which gain $K = [k_1 \quad k_2]$ and controller $u(t) = -Kx(t)$ does the closed-loop system oscillate with 1 rad/s without decaying?

- [A] $k_1 = 0, k_2 = 1$
- [B] $k_1 = 3, k_2 = 1$
- [C] For no possible K since the system is not reachable.
- [D] $k_1 = 1, k_2 = 0$

Linear Quadratic Regulator

we now know how we can place our poles at any position , but

how do we choose wisely the location of the closed loop poles?

LQR

- systematic method to trade off "tracking errors" and "control effort"
- calculates optimal K for given task
- trade off can be formulated with a cost function, which you aim to minimize

$$\begin{aligned} \min_K J(x,u) &= \int_0^\infty [x(t)^\top Q x(t) + u(t)^\top R u(t)]dt \\ \text{s.t.} \quad \dot{x}(t) &= Ax(t) + Bu(t) \\ u(t) &= -Kx(t) \end{aligned}$$

Conditions:

- Q is positive semidefinite
- R is positive definite
- $(A, \sqrt{Q})$ is detectable
- (A, B) is stabilizable

- Q and R are called **weighting matrices**
- The higher the weight, the more it is **penalized**
- $Q \uparrow \rightarrow$ **fast / accurate response**
- $R \uparrow \rightarrow$ **low input energy**
- Minimizing the cost gives the **optimal solution**

Exam Question

Question 23 *Choose the correct answer. (1 Point)*

Consider the following LTI system

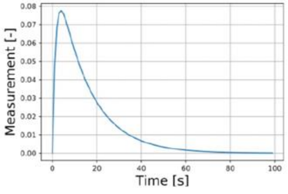
$$\dot{x}(t) = \begin{bmatrix} 0 & 1 \\ -3 & -2 \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t)$$
$$y(t) = \begin{bmatrix} 1 & 0 \end{bmatrix} x(t).$$

with associated LQR-cost functions

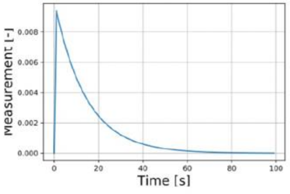
$$J_1 = \int_0^\infty x_1^2 + x_2^2 + 100u^2 dt, \qquad J_2 = \int_0^\infty 100x_1^2 + 100x_2^2 + 100u^2 dt,$$
$$J_3 = \int_0^\infty x_1^2 + x_2^2 + 0.01u^2 dt, \qquad J_4 = \int_0^\infty 100x_1^2 + 100x_2^2 + 0.01u^2 dt.$$

Consider now the following plots of the impulse responses

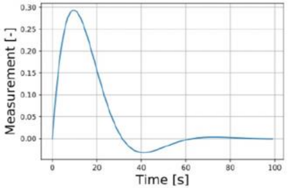
(Plot 1)



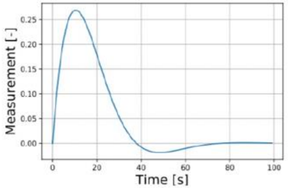
(Plot 2)



(Plot 3)



(Plot 4)



Which combination of cost functions to impulse-response plots is correct if LQR control is applied?

- [A] J_1-Plot 4, J_2-Plot 3, J_3-Plot 2, J_4-Plot 1
- [B] J_1-Plot 2, J_2-Plot 4, J_3-Plot 1, J_4-Plot 3
- [C] J_1-Plot 3, J_2-Plot 1, J_3-Plot 4, J_4-Plot 2
- [D] J_1-Plot 3, J_2-Plot 4, J_3-Plot 1, J_4-Plot 2

We have just defined a minimization problem, the next step is to solve it, so we get K for a given problem

Ricatti Equation (CARE)

- Solving the problem stated before leads to the optimal controller K
- Minimizing the problem leads to the Riccati equation:

B matrix from state-space

$$\boxed{A^\top P + PA + Q - PBR^{-1}B^\top P = 0}$$

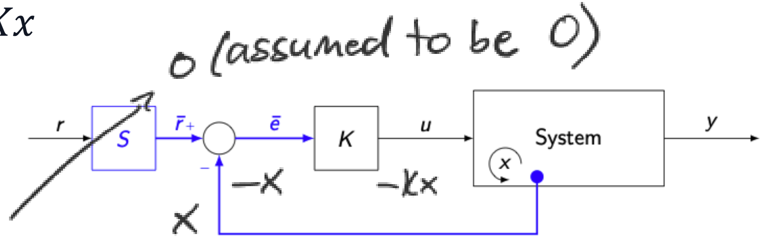
A matrix from state-space

weight matrix from cost function

→ Solve this equation for P (positive definite and symmetric!)

→ The optimal controller is calculated by: $K = R^{-1}B^\top P$

with this we can calculate the system input $u = -Kx$



- this is the continuous time solution → CARE

Ricatti Equation (DARE)

in discrete-time the ARE looks a bit different, the concept is the same:

$$\boxed{P = A^T P A - (A^T P B)(B^T P B + R)^{-1}(B^T P A) + Q}$$

- $u = -Kx$
- $K = (R + B^T P B)^{-1} B^T P A$
- P is positive definite and symmetric (just like in CT)

<u>LQR guarantees that the cl system is stable, what a phase margin of at least 60° (both CT & DT)</u>

Exam Question

$$A^\top P + PA + Q - PBR^{-1}B^\top P = 0$$

$$K = R^{-1}B^\top P$$

Question 24 *Choose the correct answer. (1 Point)*

You are given the following LTI-system:

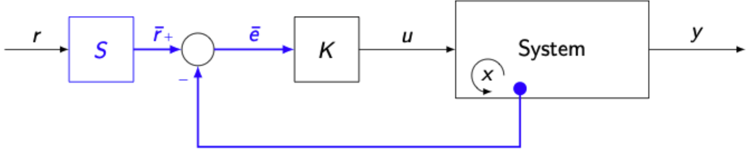
$$\dot{x}(t) = -x(t) + u(t)$$
$$y(t) = x(t)$$

with cost functions

$$J = \int_0^\infty x^2 + 0.1u^2 dt,$$

What is the LQR-gain K?

- [A] $K = -1 + \sqrt{2}$
- [B] $K = 0$
- [C] $K = -2 + \sqrt{2}$
- [D] $K = -1 + \sqrt{11}$



LQR servo

- Before we said that to achieve $e_{ss} = 0$ we need a unity gain $G_{yr}^{cl}(0) = 1$ and scaled S accordingly
- This feed-forward reference scaling is not robust ← (to disturbances and uncertainties in the plant)
- but we still want zero steady-state error AND a robust system
- remember in CS1 we just added integrators to get an $e_{ss} = 0$

▶ The idea of the LQR servo is similar: extend the state by adding the integral of the error, and then apply LQR to the extended state-space model

$$\frac{d}{dt}\begin{bmatrix} x \\ \xi \end{bmatrix} = \begin{bmatrix} A & 0 \\ -C & 0 \end{bmatrix}\begin{bmatrix} x \\ \xi \end{bmatrix} + \begin{bmatrix} B \\ 0 \end{bmatrix} u + \begin{bmatrix} 0 \\ I \end{bmatrix} r.$$

- Applying LQR to the extended system penalizes the integral of the tracking error → forces steady-state error to converge to zero.

Tips for Problem Set 2

1: We did this example together, refer to the slides. Check the reachability condition!

2: Take a closer look at reachable canonical form, for b) both forms need to have the same eigenvalues, for c) and d) use the definition for the reachability matrix, in the lecture the transformation matrix was derived as $T = \tilde{R}R^{-1}$

3: Think about what the weights Q and R mean, similar to the exam question we solved together

4: Follow linearization procedure from CS1, remember how LQR problem was defined, solve CARE, lecture slide 59 for brysons'rule. Rather a hard problem try to follow the solutions if stuck

5: a) look up formula in slides, b) is the derivation of the extended state space from slides

6: Symmetric RL has 2n branches, refer to lecture slide 72

Feedback?

Too fast? Too slow? Less theory, more exercises?

I would appreciate your feedback. Please let me know.